

HDK on DeepResearch-Bench — official-judge re-evaluation (initial report)

Date: 2026-06-29 · **Prepared for:** Zuhair · **By:** Kazim (with Claude) **Status:** initial / directional — small sample, single run. Not a leaderboard submission.

TL;DR

1. We re-scored our 2026-06-23 HDK research runs with **DeepResearch-Bench's official judge** (RACE = `openai/gpt-5.5`, FACT = `openai/gpt-5.4-mini` + Jina), so these numbers are on the **same scale as the public leaderboard** — unlike our earlier Claude-Opus-judged estimates.
 2. **HDK-4B (the default, on-device model) scored RACE 0.405** — within ~4% of the shipped **Claude-3.7-Sonnet** reference (0.422), and *matching* it on **Insight** and **Readability**. A 4B model running locally is producing frontier-adjacent report quality.
 3. **Counterintuitive but robust:** scaling the model **4B → 32B → 72B made RACE worse**, and it holds on the identical task subset (0.395 → 0.340 → 0.305). The likely driver is **report length / generation config**, not model capability (4B reports ≈16K chars vs ≈10K for 32B/72B). **This points the lever at the harness/orchestration, not bigger weights** — which is the HDK thesis.
 4. **FACT (citations) is the clear weakness, and we now know exactly why.** Two separate problems: (a) a **citation-format** issue — the 4B emits an end-of-document "Sources" list with no in-text citations, which the benchmark scores as **zero** by rule; (b) a **grounding** issue — the 32B/72B *do* cite inline, but only **14–38%** of citations actually check out, vs Claude's **87%**. Both are fixable in the synthesis step.
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Method

- **What this is:** we did **not** re-generate articles. We took the reports HDK produced on 2026-06-23/24 (Scaleway NVIDIA H100/L4) and ran them through the unmodified official benchmark harness (`Ayanami0730/deep_research_bench`, GPT-5 judge variant).
 - **Judge:** RACE = `openai/gpt-5.5` (medium reasoning); FACT = `openai/gpt-5.4-mini` + Jina page scraping. Via OpenRouter.
 - **Sample:** 5 English DeepResearch-Bench tasks (ids 51, 58, 66, 75, 83). Coverage varies by model because some runs stalled/timed out:
 - **HDK-4B** (`Qwen3.5-4B-Q4_K_M`, default) — 4 tasks (task 83 planner-stalled in the original run)
 - **HDK-32B** (`Qwen3-32B-Q4`) — 5 tasks
 - **HDK-72B** (`Qwen2.5-72B-Q4`) — 4 tasks (task 58 timed out in the original run)
 - All HDK runs = multi-agent + web (Tavily), Qwen3-Reranker-0.6B.
 - **Claude-3.7-Sonnet** = the benchmark's shipped reference output, **100 tasks (50 zh + 50 en)** — an *anchor for the judge's scale*, not a same-sample comparison.
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RACE — report quality (0–1; higher is better)

Dimension	HDK-4B	HDK-32B	HDK-72B	Claude-3.7-Sonnet (ref)
Comprehensiveness	0.383	0.324	0.284	0.411
Insight	0.401	0.367	0.276	0.405
Instruction-Following	0.430	0.366	0.331	0.462
Readability	0.418	0.367	0.362	0.417
Overall	0.405	0.354	0.306	0.422
(tasks scored)	4	5	4	100

Same-task control (tasks 51, 66, 75 — the three all models share):

	HDK-4B	HDK-32B	HDK-72B
Overall RACE (same 3 tasks)	0.395	0.340	0.305

The size-inversion is **not** a sampling artifact — it holds on identical tasks, and per-task the 4B wins or ties every shared task (appendix). The most likely cause is **report length / generation budget**: 4B reports averaged **~16.1K chars**, 32B **~10.2K**, 72B **~10.8K**, and RACE rewards comprehensiveness. The larger models, under the same agent/tool-turn budget, produced **terser** reports (and the 72B run had a timeout). **Interpretation: the bottleneck here is the harness configuration, not the weights.**

FACT — citation trustworthiness (per-report averages)

	citations found	valid citations	valid rate
HDK-4B	0	0	— (no in-text citations — see below)
HDK-32B	5.5	0.75	13.6%
HDK-72B	16.0	6.0	37.5%
Claude-3.7-Sonnet	28.1	24.5	87.3%

Two distinct problems, both in the synthesis step:

1. **Format (the 4B's zero).** The benchmark's extractor only counts citations that are **placed in-text** at the claim. Its rule, verbatim: *"If the main text does not specify the exact location of the citation (e.g. only a reference list at the end of the article, without specifying the citation point in the text), please return an empty list."* The 4B writes claims in prose, then a `## Sources` list at the end with bare URLs — so it scores **0**, even though the URLs exist.
2. **Grounding (the 32B/72B low validity).** When HDK does cite inline, only **14–38%** of those citations are actually supported by the page they point to (we observed fabricated/altered URLs), vs Claude's **87%**.

What this means

- **The on-device research engine is real and competitive on quality.** A laptop-class 4B at RACE 0.405 — neck-and-neck with Claude-3.7-Sonnet on Insight and Readability — is a strong "engine > size" data point on the *official* scale.
- **Bigger ≠ better here.** Scaling to 32B/72B *reduced* RACE under the current harness config. That's a strong signal that **investment should go into the orchestration/synthesis (report depth, agent budget, no timeouts), not larger weights**. It also means the cheap, distributable 4B is the right product default — confirmed, not assumed.
- **Citations are the gap to close before any public leaderboard entry**, and the fix is now precise: (1) make synthesis emit **in-text** citations ([Title](url) at the claim, not an end list) — this alone takes the 4B off zero; (2) **constrain cited URLs to the actually-fetched source set** and validate them — this lifts the 14–38% validity toward Claude's 87%.

Caveats (read these)

- **Tiny sample:** 3–5 English tasks, **single run, no variance** — directional only.
- **Unequal coverage:** model sub-samples differ (4B lacks task 83, 72B lacks task 58); the same-task control mitigates but doesn't eliminate this.
- **Length confound:** the cross-size RACE gap is entangled with report length / generation config and needs a controlled re-run to attribute cleanly.
- **English-only:** the 4B's Chinese (50% of the real benchmark) is untested.
- **Claude column** is the shipped 100-task reference, not run on our tasks — a scale anchor, not a head-to-head.

Recommended next steps

1. **Controlled re-run** — same tasks, same fixed generation config (findings budget, agent count, `n-ctx`, no timeouts) across 4B/32B/72B, to cleanly isolate the model-size effect. Hypothesis: with matched report length, the size gap shrinks and the 4B remains the sweet spot.
2. **Citation fix, two steps** — (a) in-text citation format in the synthesis prompt; (b) ground cited URLs to the fetched source set + validate. Re-measure FACT on this same 5-task set to confirm uplift.
3. **Then scale** — once citations are non-zero and grounded, run a full 50-EN (or 100-task) pass on the chosen config for a real, submittable leaderboard number.

Appendix — per-task overall RACE

task	HDK-4B	HDK-32B	HDK-72B
51	0.377	0.293	0.268
58	0.436	0.396	—
66	0.390	0.345	0.361
75	0.417	0.381	0.287
83	—	0.355	0.307

Reproduce: `code/my-work/deep_research_bench/` — raw articles in `data/test_data/raw_data/hdk-*.jsonl`, results in `results/race/<model>/` and `results/fact/<model>/`. Re-run: fill `keys.env.local`, then `./run-hdk-eval.sh`. Source reports: `planning/specs/nvidia-benchmark-2026-06-23/raw-results*.tgz`.